Solutions and discussion for Assignment 4b.

Question 1. We are given the effective Hamiltonian in the flavor basis as,

$$\hat{H} = U \begin{pmatrix} E_1^0 & 0 \\ 0 & E_2^0 \end{pmatrix} U^\dagger + \begin{pmatrix} V_{CC} & 0 \\ 0 & 0 \end{pmatrix}. \tag{1.3}$$

As mentioned in the hint, terms proportional to the identity matrix will not contribute to oscillations. This means we can subtract any term proportional to the identity matrix from the Hamiltonian without changing the oscillation physics. Therefore, we can write:

$$\begin{aligned} U \begin{pmatrix} E_1^0 & 0 \\ 0 & E_2^0 \end{pmatrix} U^\dagger &= U \left[\frac{E_1^0 + E_2^0}{2} \mathbb{1} + \frac{E_2^0 - E_1^0}{2} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \right] U^\dagger & (1.4) \\ &= \frac{E_1^0 + E_2^0}{2} \mathbb{1} + \frac{\Delta m^2}{4E_\nu} U \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} U^\dagger & (1.5) \\ &= p \times \mathbb{1} + \frac{\Delta m^2}{4E_\nu} \begin{pmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}, & (1.6) \end{aligned}$$

where we used $E_i^0 \simeq p + \frac{m_i^2}{2p}$, and will drop the p term and replace $p \to E_\nu$. Adding the matter potential term, we have

$$\hat{H} \simeq \frac{\Delta m^2}{4E_\nu} \begin{pmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix} + \begin{pmatrix} V_{CC} & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} V_{CC} - \frac{\Delta m^2}{4E_\nu} \cos 2\theta & \frac{\Delta m^2}{4E_\nu} \sin 2\theta \\ \frac{\Delta m^2}{4E_\nu} \sin 2\theta & \frac{\Delta m^2}{4E_\nu} \cos 2\theta \end{pmatrix}. \tag{1.7}$$

Subtracting $\frac{V_{CC}}{2}\mathbb{1}$ from the Hamiltonian (which does not affect oscillations), we have

$$\boxed{\hat{H} \simeq \frac{\Delta m^2}{4E_\nu} \begin{pmatrix} r - \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -r + \cos 2\theta \end{pmatrix}, \text{ where } r = \frac{A_{\mathrm{CC}}}{\Delta m^2} \text{ and } A_{\mathrm{CC}} \equiv 2E_\nu V_{CC}.} \tag{1.8}$$

We can now use the hint about diagonalizing a 2×2 matrix to write

$$\tan 2\theta_m = \frac{\tan 2\theta}{1 - \frac{1}{r \cos 2\theta}}. \tag{1.9}$$

Using the identity $\sin 2\theta_m = \tan 2\theta_m / \sqrt{1 + \tan^2 2\theta_m}$ for the quadrant $0 < \theta < \pi/4$[1], we can use

$$\boxed{\sin^2 2\theta_m = \frac{\sin^2 2\theta}{(\cos 2\theta - r)^2 + \sin^2 2\theta}, \quad \text{where} \quad r \equiv \frac{A_{\mathrm{CC}}}{\Delta m^2}.} \tag{1.10}$$

For completeness, we can also write

$$\boxed{\cos^2 2\theta_m = \frac{(\cos 2\theta - r)^2}{(\cos 2\theta - r)^2 + \sin^2 2\theta}, \quad \text{where} \quad r \equiv \frac{A_{\mathrm{CC}}}{\Delta m^2}.} \tag{1.11}$$

[1]This choice is correct for θ_{12} and θ_{13}, but experimentally we still do not know if it is the case for θ_{23}, which is either a little larger or smaller than $\pi/4$.

The eigenvalues can be found to be

$$\boxed{E^m_{1,2} = \mp \frac{\Delta m^2}{4E_\nu}\sqrt{1 + r^2 - 2r\cos 2\theta}.} \tag{1.12}$$

The mass splitting in matter is then

$$\Delta m^2_m = 2E_\nu(E^m_2 - E^m_1) = \Delta m^2 \sqrt{(\cos 2\theta - r)^2 + \sin^2 2\theta}. \tag{1.13}$$

As desired, we have the form $\sin^2 2\theta_m = \sin^2 2\theta / R$ and $\Delta m^2_m = \Delta m^2 \sqrt{R}$, where

$$\boxed{R = (\cos 2\theta - r)^2 + \sin^2 2\theta}. \tag{1.14}$$

To rotate to the matter eigenbasis (the basis where the Hamiltonian is diagonal in the presence of the potential term), we use the mixing matrix with angle θ_m, i.e., $U_m = R(\theta_m)$, so that

$$\boxed{\nu_m = U^\dagger_m \nu_{\text{flavor}}}, \text{ where } \nu_m = \begin{pmatrix} \nu^m_1 \\ \nu^m_2 \end{pmatrix} \text{ and } \nu_{\text{flavor}} = \begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix}. \tag{1.15}$$

Let us carefully analyse what happens to Schrodinger's equation as we rotate to the matter eigenbasis,

$$\begin{aligned} i\frac{d}{dt}\nu_{\text{flavor}} = \hat{H}\nu_{\text{flavor}} &\implies i\frac{d}{dt}(U_m\nu_m) = \hat{H}(U_m\nu_m). \\ &\implies iU_m\frac{d}{dt}\nu_m + i\frac{dU_m}{dt}\nu_m = \hat{H}U_m\nu_m. \\ &\implies \boxed{i\frac{d}{dt}\nu_m = U^\dagger_m\hat{H}U_m\nu_m - iU^\dagger_m\frac{dU_m}{dt}\nu_m.} \end{aligned} \tag{1.16}$$

For the moment, we are assuming that the matter potential is constant, so that U_m is also constant and $\frac{dU_m}{dt} = 0$. Therefore, we have

$$i\frac{d}{dt}\nu_m = U^\dagger_m\hat{H}U_m\nu_m = \begin{pmatrix} E^m_1 & 0 \\ 0 & E^m_2 \end{pmatrix}\nu_m. \tag{1.17}$$

In the Bonus Question, we will consider the case where the matter potential is changing, so we will need to keep track of the $\frac{dU_m}{dt}$ term in Eq. (1.16).

Note that the matter eigenstates ν_i^m are not the same as the mass eigenstates in vacuum ν_i. The matter eigenstates diagonalize the Hamiltonian in the presence of the matter potential, while the vacuum mass eigenstates diagonalize the Hamiltonian in vacuum. The relation between the flavor eigenstates and the vacuum mass eigenstates is still the same, except that in matter, the oscillation probability is most easily calculated in the matter eigenbasis using the effective mixing angle θ_m and mass splitting Δm_m^2 derived above. By complete analogy with the vacuum case, in the *constant density* case, the oscillation probability in matter is given by

$$P(\nu_e \to \nu_\mu) = \sin^2 2\theta_m \sin^2\left(\frac{\Delta m_m^2 t}{4E_\nu}\right) = \frac{\sin^2 2\theta}{R} \sin^2\left(\frac{\Delta m^2 t}{4E_\nu}\sqrt{R}\right). \tag{1.18}$$

Question 2. The effective mixing angle in matter becomes maximal when $\sin^2 2\theta_m = 1$, which occurs when the denominator is minimized, i.e., when

$$r \overset{!}{=} \cos 2\theta \implies A_{\mathrm{CC}} = \Delta m^2 \cos 2\theta \text{ and } R = \sin^2 2\theta. \tag{1.19}$$

Here I use $\overset{!}{=}$ to indicate that this is the condition we are *imposing*. This is the MSW resonance condition. When this condition is satisfied, the mixing angle in matter becomes $\theta_m = \pi/4$, meaning that the flavor eigenstates are equal mixtures of the matter eigenstates:

$$\nu_e = \frac{1}{\sqrt{2}}(\nu_1^m + \nu_2^m), \quad \nu_\mu = \frac{1}{\sqrt{2}}(-\nu_1^m + \nu_2^m). \tag{1.20}$$

Because of the maximal mixing above, oscillations are maximal, i.e., the probability of oscillating from one flavor to another is maximized. To see this, we can plug the resonance condition into the oscillation probability in matter derived above to find

$$P(\nu_e \to \nu_\mu) = \sin^2\left(\frac{\Delta m^2 t}{4E_\nu}\sin 2\theta\right). \tag{1.21}$$

Note that we can get large oscillation probabilities even if the vacuum mixing angle θ is small, as long as the resonance condition is satisfied and as long as we wait for long enough time t for oscillations to occur. The only effect of a small mixing angle in the constant density MSW resonance is to reduce the effective mass splitting in matter, which increases the oscillation length.

It is instructive to plot the resonance factor R as a function of the matter potential parameter $r = A_{\mathrm{CC}}/\Delta m^2$ for different vacuum mixing angles θ. This is shown in Fig. 1. We can see that for positive r values, corresponding to neutrinos in normal ordering ($V_{\mathrm{CC}} > 0$ and $\Delta m^2 > 0$) or antineutrinos[2] in inverted ordering ($V_{\mathrm{CC}} < 0$ and $\Delta m^2 < 0$), the resonance condition can be satisfied for small mixing angles, leading to a significant reduction in the resonance factor R and thus an enhancement in the effective mixing angle θ_m. For negative r values, corresponding to antineutrinos in normal ordering ($V_{\mathrm{CC}} < 0$ and $\Delta m^2 < 0$)

[2] For antineutrinos, the sign of the matter potential is reversed, $V_{\mathrm{CC}} < 0$.

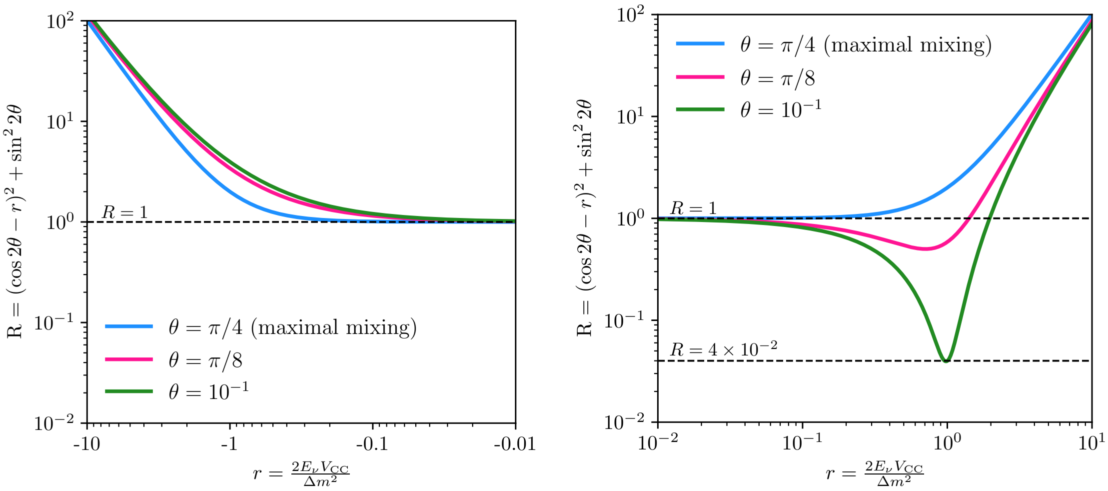

Figure 1: The resonance factor $R = (\cos 2\theta - r)^2 + \sin^2 2\theta$ as a function of $r = A_{\rm CC}/\Delta m^2$ for different vacuum mixing angles θ. Left: negative r values (antineutrinos in normal ordering or neutrinos in inverted ordering). Right: positive r values (neutrinos in normal ordering or antineutrinos in inverted ordering).

or neutrinos in inverted ordering ($V_{\rm CC} > 0$ and $\Delta m^2 > 0$), the resonance condition cannot be satisfied for small mixing angles, and the resonance factor R remains relatively large, indicating that the effective mixing angle θ_m does not become maximal.

Question 3. The MSW resonance condition is given by $A_{\rm CC} = \Delta m^2 \cos 2\theta$. Using $A_{\rm CC} = 2E_\nu V_{CC} = 2E_\nu\sqrt{2}G_F N_e$, we can solve for the neutrino energy at which the resonance occurs:

$$E_\nu^{\rm res} = \frac{\Delta m^2 \cos 2\theta}{2\sqrt{2}G_F N_e}. \tag{1.22}$$

Plugging in the numbers, we have

$$E_\nu^{\rm res} \simeq \frac{7.5 \times 10^{-5}\ \mathrm{eV}^2 \times \cos 66^\circ}{2\sqrt{2} \times 1.166 \times 10^{-5}\ \mathrm{GeV}^{-2} \times 100 \times 6.022 \times 10^{23}\ \mathrm{cm}^{-3}} \simeq 2\ \mathrm{MeV}. \tag{1.23}$$

This energy is within the range of solar neutrino energies, indicating that solar neutrinos can experience the MSW resonance as they propagate through the Sun. In reality, the electron density in the Sun varies with radius, so solar neutrinos can experience a range of matter potentials as they propagate outward from the core to the surface of the Sun. The actual resonance happens at a specific radius where the local electron density satisfies the resonance condition for a given neutrino energy, which turns out to be a bit higher than 2 MeV for typical solar neutrinos.

Question 4. When the MSW resonance condition is exactly satisfied throughout the entire neutrino propagation, we have $\sin^2 2\theta_m = 1$ and $\Delta m_m^2 = \Delta m^2 \sin 2\theta$. For $\Delta m^2 t/2E_\nu \gg 1$, the argument of the sine function in the oscillation probability becomes very large, leading to rapid oscillations. In this case, the oscillation probability averages out to

$$\langle P(\nu_e \to \nu_\mu)\rangle = \frac{1}{2}. \tag{1.24}$$

This means that there is a 50% chance of finding the neutrino in either flavor state after propagation. For $\Delta m_m^2 t/2E_\nu \ll 1$, we can expand the sine function to first order:

$$\sin^2\left(\frac{\Delta m_m^2 t}{4E_\nu}\right) \simeq \left(\frac{\Delta m_m^2 t}{4E_\nu}\right)^2. \tag{1.25}$$

Plugging this into the oscillation probability, we find

$$P(\nu_e \to \nu_\mu) \simeq \sin^2 2\theta_m \left(\frac{\Delta m_m^2 t}{4E_\nu}\right)^2 = \left(\frac{\Delta m^2 t}{4E_\nu}\sin 2\theta\right)^2. \tag{1.26}$$

In this regime, the oscillation probability is suppressed and depends quadratically on the small parameter $\Delta m^2 t/2E_\nu$. Note that the formula looks exactly what you would get for *early* oscillations in the vacuum case.

In fact, this is a more general result that holds even when the MSW resonance condition is not exactly satisfied. For instance, consider the case where the oscillation phase is small, i.e., $\Delta m_m^2 t/2E_\nu\sqrt{R} \ll 1$. In that case, we can expand the sine function in the oscillation probability to find

$$P(\nu_e \to \nu_\mu) \simeq \sin^2 2\theta_m \left(\frac{\Delta m_m^2 t}{4E_\nu}\right)^2 = \frac{\sin^2 2\theta}{R}\left(\frac{\Delta m^2 t}{4E_\nu}\sqrt{R}\right)^2 = \sin^2 2\theta\left(\frac{\Delta m^2 t}{4E_\nu}\right)^2. \tag{1.27}$$

which is again the same as the vacuum case for early oscillations. This is called *vacuum mimicking*:

$$\boxed{P(\nu_e \to \nu_\mu)_{\text{matter}} \xrightarrow[\text{oscillation}]{\text{early}} P(\nu_e \to \nu_\mu)_{\text{vacuum}} = \sin^2 2\theta\left(\frac{\Delta m^2 t}{4E_\nu}\right)^2.} \tag{1.28}$$

Bonus Question 5. The goal of this question was to make you read more about the MSW effect and think about the case where the neutrino experiences a changing matter density along its trajectory. This is case for solar neutrinos propagating from the core of the Sun to its surface, as well as for supernova neutrinos propagating through the outer layers of the star. In this case, the matter potential V_{CC} is a function of position (or time, since we are taking $t \simeq x$), and so the effective mixing angle θ_m also becomes a function of time. To analyze this case, we need to keep track of the $\frac{dU_m}{dt}$ term in Eq. (1.16) derived in the infobox above.

Let us include the time-derivative of the rotation using $U_m = R(\theta_m)$ and

$$\frac{dU_m}{dt} = \frac{d\theta_m}{dt}\begin{pmatrix} -\sin\theta_m & \cos\theta_m \\ -\cos\theta_m & -\sin\theta_m \end{pmatrix}. \tag{1.29}$$

Plugging this into Eq. (1.16), you can show that the time evolution in the matter eigenbasis simplifies to

$$i\frac{d}{dt}\nu_m = \begin{pmatrix} E_1^m & -i\frac{d\theta_m}{dt} \\ i\frac{d\theta_m}{dt} & E_2^m \end{pmatrix} \nu_m. \tag{1.30}$$

While before we could time evolve the matter eigenbasis using $|\nu_m(t)\rangle = e^{-iE_m t}|\nu_m\rangle$, now this is no longer true because the off-diagonal terms proportional to $\frac{d\theta_m}{dt}$ can induce transitions between the matter eigenstates ν_1^m and ν_2^m.

Let us investigate the derivative term a bit more. Using the expression for $\sin 2\theta_m$ derived in Question 1, we can write

$$\frac{d\theta_m}{dx} = \frac{1}{2}\frac{d}{dx}(\arcsin\sin 2\theta_m) = \frac{1}{2}\frac{1}{\sqrt{1-\sin^2 2\theta_m}}\frac{d}{dx}(\sin 2\theta_m) = \frac{1}{2}\frac{1}{\cos 2\theta_m}\frac{d}{dx}(\sin 2\theta_m). \tag{1.31}$$

Using Eq. (1.10), we find

$$\frac{d}{dx}(\sin 2\theta_m) = \sin 2\theta \frac{d}{dx}\left[\frac{1}{R}\right] = -\sin 2\theta \frac{(\cos 2\theta - r)}{R^{3/2}}\frac{dr}{dx}. \tag{1.32}$$

hence using $\sin 2\theta_m = \sin 2\theta/\sqrt{R}$ and $\Delta m_m^2 = \Delta m^2\sqrt{R}$, as well as $\cos 2\theta_m = (\cos 2\theta - r)/\sqrt{R}$, we have

$$\frac{d\theta_m}{dx} = \frac{1}{2}\frac{\sin^2 2\theta_m}{\Delta m_m^2}\frac{(\cos 2\theta - r)}{\sqrt{R}\cos 2\theta_m}\frac{dA_{\mathrm{CC}}}{dx} = \frac{1}{2}\frac{\sin^2 2\theta_M}{\Delta m_M^2}\frac{dA_{\mathrm{CC}}}{dx}. \tag{1.33}$$

So, unsurprisingly, the derivative of the mixing angle in matter is proportional to the derivative of the matter potential. If the matter potential varies slowly, i.e., if $\frac{dA_{\mathrm{CC}}}{dx}$ is small, then $\frac{d\theta_m}{dx}$ will also be small.

The adiabaticity condition requires that the off-diagonal terms in the Hamiltonian in the matter eigenbasis are much smaller than the diagonal terms. In practice, this means that the for each time step Δt, the change in the mixing angle $\Delta\theta_m$ should be much smaller than the difference in the eigenvalues multiplied by the time step, i.e.,

$$\left|\frac{d\theta_m}{dt}\Delta t\right| \ll |E_2^m - E_1^m|\Delta t. \tag{1.34}$$

Dividing both sides by Δt and rearranging, we find the adiabaticity condition:

$$\left|\frac{d\theta_m}{dt}\right| \ll |E_2^m - E_1^m| = \frac{\Delta m_m^2}{2E_\nu}. \tag{1.35}$$

You will often see this condition written in terms of the adiabaticity parameter γ defined as

$$\boxed{\gamma \equiv \frac{\Delta m_m^2/2E_\nu}{|d\theta_m/dt|} = \frac{(\Delta m_m^2)^2}{\sin^2 2\theta_m E_\nu |dA_{\mathrm{CC}}/dt|} \gg 1.} \tag{1.36}$$

When this condition is satisfied, the time evolution of the neutrino state in matter is adiabatic, meaning that for a small time step Δt, the neutrino will not want to jump from one matter eigenstate to another.

Accounting for a time dependence of $E^m_{1,2}$, the two differential equations to be solved to find the time evolution of a neutrino state. Identifying the time coordinate with the distance traveled by the neutrino $t = x$, we get

$$i\frac{d}{dx}\nu^m_1 = E^m_1\nu^m_1 - i\frac{d\theta_m}{dx}\nu^m_2, \quad i\frac{d}{dx}\nu^m_2 = i\frac{d\theta_m}{dx}\nu^m_1 + E^m_2\nu^m_2. \tag{1.37}$$

In a small time step dx, we have

$$\begin{aligned}\nu^m_1(x+dx) &\simeq (1 - iE^m_1 dx)\nu^m_1(x) - \frac{d\theta_m}{dx}dx\,\nu^m_2(x),\\ \nu^m_2(x+dx) &\simeq (1 - iE^m_2 dx)\nu^m_2(x) + \frac{d\theta_m}{dx}dx\,\nu^m_1(x).\end{aligned} \tag{1.38}$$

As long as the derivative term $\frac{d\theta_m}{dx}$ is small compared to the diagonal terms E^m_1 and E^m_2, the neutrino will remain in the same matter eigenstate throughout the propagation. This is the key idea behind the adiabaticity condition.

Neglecting the derivative term $\frac{d\theta_m}{dt}$ altogether, the matter eigenstates evolve independently with phases $|\nu^m_i(L)\rangle = e^{-i\int_0^L E^m_i dx}|\nu^m_i(0)\rangle$. However, at the production and detection points, the rotation from flavor to matter basis is, in general, different, since the matter potential (and thus the mixing angle θ_m) is different at these two locations. The flavor states can be related to the matter eigenbasis at the production point $x = 0$ or the detection point $x = L$. In fact, using the fact that the matter eigenstates evolve independently in the adiabatic limit, we have that

$$|\nu_\alpha\rangle = \sum_i U^{m*}_{\alpha i}(0)\,|\nu^m_i(0)\rangle = \sum_i U^{m*}_{\alpha i}(L)\,|\nu^m_i(L)\rangle\,, \tag{1.39}$$

$$|\nu_\beta\rangle = \sum_i U^{m*}_{\beta i}(L)\,|\nu^m_i(L)\rangle = \sum_i U^{m*}_{\beta i}(L)e^{-i\int_0^L E^m_i dx}\,|\nu^m_i(0)\rangle\,, \tag{1.40}$$

where $U^m_{\alpha i}(0)$ and $U^m_{\beta i}(L)$ are the elements of the matter mixing matrix U_m evaluated with mixing angles at the production and detection points, respectively. Putting these two facts together, if adiabaticity is satisfied, we can write the flavor transformation amplitude as

$$\begin{aligned}A(\nu_\alpha\to\nu_\beta) = \langle\nu_\beta|\nu_\alpha(L)\rangle &= \left(\sum_j \langle\nu^m_j(0)|\,U^m_{\beta j}(0)\right)\left(\sum_i e^{-i\int_0^L E^m_i dx}U^{m*}_{\alpha i}(0)\,|\nu^m_i(0)\rangle\right).\\ &= \sum_i U^m_{\beta i}e^{-i\int_0^L E^m_i dx}U^{m*}_{\alpha i}(0).\end{aligned} \tag{1.41}$$

where the integral in the exponential is over the position-dependent matter eigenvalues $E^m_i(x)$. Note that we chose to write the bra in terms of the matter eigenstates at the production point $x = 0$ in order to force the kets to also be at $x = 0$ for the inner product to be well-defined. This way, we have explicitly made use of the time evolution of the matter eigenstates from $x = 0$ to $x = L$.

The probability is then given by $P(\nu_\alpha \to \nu_\beta) = |A(\nu_\alpha \to \nu_\beta)|^2$, which can be computed once the matter density profile (and thus $E_i^m(x)$ and $U_m(x)$) is known. Explicitly,

$$P(\nu_\alpha \to \nu_\beta) = \left| \sum_i U_{\beta i}^m(L) e^{-i\int_0^L E_i^m dx} U_{\alpha i}^{m*}(0) \right|^2 . \tag{1.42}$$

$$= \sum_{i,j} U_{\beta i}^m(L) U_{\beta j}^{m*}(L) U_{\alpha i}^{m*}(0) U_{\alpha j}^m(0) e^{-i\int_0^L (E_i^m - E_j^m)dx} \tag{1.43}$$

$$= \sum_i |U_{\beta i}^m(L)|^2 |U_{\alpha i}^m(0)|^2 + 2\sum_{i>j} \mathrm{Re}\left[U_{\beta i}^m(L) U_{\beta j}^{m*}(L) U_{\alpha i}^{m*}(0) U_{\alpha j}^m(0) e^{-i\Delta_{ij}(L)} \right] \tag{1.44}$$

$$= \sum_i |U_{\beta i}^m(L)|^2 |U_{\alpha i}^m(0)|^2 + 2\sum_{i>j} \mathrm{Re}\left[U_{\beta i}^m(L) U_{\beta j}^{m*}(L) U_{\alpha i}^{m*}(0) U_{\alpha j}^m(0) \right] \cos\left(\Delta_{ij}(L)\right) \tag{1.45}$$

$$+ 2\sum_{i>j} \mathrm{Im}\left[U_{\beta i}^m(L) U_{\beta j}^{m*}(L) U_{\alpha i}^{m*}(0) U_{\alpha j}^m(0) \right] \sin\left(\Delta_{ij}(L)\right), \tag{1.46}$$

where we defined the phase difference

$$\Delta_{ij}(L) = \int_0^L (E_i^m - E_j^m) dx = \int_0^L \frac{\Delta m_m^2(x)}{2E_\nu} dx. \tag{1.47}$$

Take for simplicity a two-flavor system and consider the disappearance channel $\nu_e \to \nu_e$. In that case, we have

$$P(\nu_e \to \nu_e) = |U_{e1}^m(L)|^2 |U_{e1}^m(0)|^2 + |U_{e2}^m(L)|^2 |U_{e2}^m(0)|^2 \tag{1.48}$$
$$+ 2\,\mathrm{Re}\,[U_{e1}^m(L) U_{e2}^{m*}(L) U_{e1}^{m*}(0) U_{e2}^m(0)] \cos\left(\Delta_{21}(L)\right)$$
$$= \cos^2\theta_m(L)\cos^2\theta_m(0) + \sin^2\theta_m(L)\sin^2\theta_m(0) \tag{1.49}$$
$$+ 2\cos\theta_m(L)\sin\theta_m(L)\cos\theta_m(0)\sin\theta_m(0)\cos\left(\Delta_{21}(L)\right).$$

Rearranging, we find

$$P(\nu_e \to \nu_e) = \frac{1}{2}\left[1 + \cos 2\theta_m(L)\cos 2\theta_m(0) + \sin 2\theta_m(L)\sin 2\theta_m(0)\cos\left(\int_0^L dx \frac{\Delta m_m^2(x)}{2E_\nu} \right) \right]. \tag{1.50}$$

This is highly significant. The oscillation probability depends on the mixing angles at production and detection, but not in-between. Indeed, the only dependence on the history of the neutrino comes in through the oscillatory cosine term. This usually is very hard to observed because the cases for which matter effects are relevant usually involve very far away sources like the Sun and supernovae. In those cases the cosine term averages out to 0 and we are left with the probability

$$P(\nu_e \to \nu_e) = |U_{e1}^m(L)|^2 |U_{e1}^m(0)|^2 + |U_{e2}^m(L)|^2 |U_{e2}^m(0)|^2 = \frac{1}{2}\left[1 + \cos 2\theta_m(L)\cos 2\theta_m(0)\right]. \tag{1.51}$$

where only the mixing angles at production and detection matter.

Another important point to note is that when the neutrino is born in a very high density environment (above the MSW resonance, $R \gg 1$) and is detected in a low density environment (below the MSW resonance, $R \simeq 1$), the mixing angles change significantly, going from $\theta_m(0) \simeq \pi/2$ to $\theta_m(L) \simeq \theta$. This means that if the neutrino is produced as a pure ν_e flavor state at high density, it is almost purely the heavier matter eigenstate ν_2^m:

$$|\nu_e\rangle = \cos\theta_m(0)\,|\nu_1^m\rangle + \sin\theta_m(0)\,|\nu_2^m\rangle \simeq |\nu_2^m\rangle\,. \tag{1.52}$$

If the neutrino then propagates adiabatically to low density, it will remain in the $\nu_2^m(x)$ state throughout its journey all the way out of the matter region, when it finds itself in the ν_2 state, the vacuum mass eigenstate:

$$|\nu_2^m(L)\rangle \simeq |\nu_2\rangle\,. \tag{1.53}$$

That is, an adiabatic propagation through a changing density profile maintains the $\nu_e \sim \nu_2^m(0)$ state produced at high density until it finds itself in the $\nu_2(L) \sim \nu_2$ vacuum mass eigenstate at low density.

We can see this explicitly in the probability. Using the fact that $\sqrt{R} \simeq r$ in the high density region ($R \gg 1$) and that $\cos 2\theta_m(0) \simeq (\cos 2\theta - r)/r \simeq -1$. Plugging this into the averaged oscillation probability above, we find

$$\begin{aligned} P(\nu_e \to \nu_e) &= \frac{1}{2}\left[1 + \cos 2\theta_m(L)\cos 2\theta_m(0)\right] \\ &= \frac{1}{2}\left[1 + \cos 2\theta \cos 2\theta_m(0)\right]. \\ &= \frac{1}{2}\left[1 + \cos 2\theta \frac{(\cos 2\theta - r)}{\sqrt{R}}\right] \\ &\simeq \frac{1}{2}\left[1 - \cos 2\theta\right] = \sin^2\theta = |U_{e2}|^2. \end{aligned} \tag{1.54}$$

The fact that the probability is just $|U_{e2}|^2$ means that to detect the neutrino as a ν_e flavor state we must project the ν_2 mass eigenstate onto the ν_e flavor state, which gives exactly $|U_{e2}|^2$.

Therefore, solar neutrinos produced as ν_e in the core of the Sun at very high density will exit the Sun as almost pure ν_2 mass eigenstates, provided that the propagation is adiabatic. The condition that $R \gg 1$ at production is satisfied for solar neutrinos with energies above a 4-6 MeV, approximately (this is equivalently to saying that neutrinos below this energy do not experience the MSW resonance in the Sun). This explains the solar neutrino problem (which we will hear more about from Group 0!): solar neutrinos produced as ν_e oscillate into ν_2 mass eigenstates while propagating through the Sun, and since ν_2 has only a small ν_e component ($|U_{e2}|^2 = \sin^2\theta_{12} \simeq 0.3$), the flux of ν_e detected on Earth is significantly reduced compared to the flux produced in the Sun.